

Assignment 1 Code Analysis:

- A. I used Google Gemini and Anthropic Claude
- B. I accessed Gemini through Google search (with SearchAI) and I went to Claudes website.
- C. Prompt: Write a C program that picks a secret number between 1 and 10, the user has up to 3 attempts to guess the number. The requirements include using a fixed secret number, using printf to prompt the user and scanf to read their guess, use a while or for loop to let the user guess up to 3 times, use if else statements to check if the guess is too high, low, or correct, and stop the loop immediately if the user guesses the correct number. The program will print either a win or lose message. An example format would look like:

Guess a number between 1 and 10.

Attempt 1/3. Enter your guess: 5

Too low! Try again.

Attempt 2/3. Enter your guess: 8

Too high! Try again.

Attempt 3/3. Enter your guess: 7

Correct! You win!

D.

- a. This is Claudes code:

```
#include <stdio.h>

int main() {
    int secret = 7;
    int guess;
    int attempts = 3;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= attempts; i++) {
        printf("Attempt %d/%d. Enter your guess: ", i,
attempts);
        scanf("%d", &guess);
```

```
        if (guess == secret) {
            printf("Correct! You win!\n");
            won = 1;
            break;
        } else if (guess < secret) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }
    }

    if (!won) {
        printf("Out of attempts! The number was %d. You
lose.\n", secret);
    }

    return 0;
}
```

gg. This is Gemini's code:

```
#include <stdio.h>

int main() {
    // Requirements: Fixed secret number and maximum of 3
    tries

    const int SECRET_NUMBER = 7;
    const int MAX_ATTEMPTS = 3;
    int guess;
    int win = 0; // Flag to track if the user won

    printf("Guess a number between 1 and 10.\n");

    // Loop for up to 3 attempts
    for (int attempt = 1; attempt <= MAX_ATTEMPTS;
attempt++) {
```

```
        printf("Attempt %d/%d. Enter your guess: ",
attempt, MAX_ATTEMPTS);
        scanf("%d", &guess);

        // Check the user's guess
        if (guess == SECRET_NUMBER) {
            printf("Correct! You win!\n");
            win = 1;
            break; // Stop the loop immediately on
correct guess
        } else if (guess < SECRET_NUMBER) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }
    }

    // Print final lose message if the user ran out of
tries
    if (!win) {
        printf("Game over! You lost. The correct number
was %d.\n", SECRET_NUMBER);
    }

    return 0;
}
```

- E. Claude's code doesn't include comments while Gemini does, having a better maintainability on what the program file does. Also Gemini has good compile time for adding const to variables that won't be reassigned later. Gemini has more code correctness than Claude. In terms of space and time complexity, both are $O(1)$ in time complexity since it only loops 3 times. Same thing with space complexity being $O(1)$ although they

could've used less variables, having a win and max_attempts variable seems unnecessary.

- F. For assignment 1, I am choosing Googles Gemini code for it having multiple comments that tell the user what each main block of code does, and also it has compile time safety where it prevents the programmer from reassigning variables like SECRET_NUMBER or MAX_ATTEMPTS, finally the variables being all caps makes it easier for the programmer to know they are const variables that can't be reassigned.
- G. I would add more comments on the multiple if else statements that tell if the user guessed high, low, or they were correct. I would also remove the win variable and just put an else statement outside of the for loop.